

Priyadarshini College of Engineering, Nagpur
Sessional Examination (2023-24) Even Semester
B. Tech. Eighth Semester (Computer Technology) (C. B. C. S.)
Blockchain and its Application

P. Pages: 2

Time: Three Hours

PCE/KS/24/BECT802T-2

Max. Marks: 70

Notes:

- 1) All questions carry marks as indicated.
- 2) Solve Question 1 or Question 2
- 3) Solve Question 3 or Question 4
- 4) Solve Question 5 or Question 6
- 5) Solve Question 7 or Question 8
- 6) Solve Question 9 or Question 10
- 7) Due credit will be given to neatness and adequate dimensions.
- 8) Assume suitable data wherever necessary.
- 9) Illustrate your answers whenever necessary with the help of neat sketches.

Q. No.	Questions	CO	BL	Marks
1	a) Discuss SHA256 Algorithm with processing.	CO1	2	7
	b) What is Decentralization? Explain with the help of supply chain management examples.		2	7
	OR			
2	a) Explain Cryptocurrencies and its evolution	CO1	2	7
	a) Explain Markle tree and structure of Markle tree		2	7
3	a) What are Bitcoin scripts? Explain with example	CO2	2	7
	b) What are the steps for joining a bitcoin network? Explain.		2	7
	OR			
4	a) What is double Spending? Explain how double spending is handled using blockchain.	CO2	2	7
	b) What are smart Contracts. Explain various types of smart contracts		2	7
5	a) Discuss	CO3	2	8
	I. Proof of Work(PoW) II. Proof of Stack			
	II. Proof of Burn IV. Proof of Elapsed Time			
	b) What is block mining? describe the role of miners in detail.		2	6
	OR			

6	a) Differentiate Permissioned and Permission less Blockchain b) What are the various Faults in a Distributed Systems? Explain	CO3	2 2	8 6
7	a) Discuss I. Ethereum Smart Contracts II. Ethereum Virtual Machine - EVM b) Draw and explain hyperledger fabric architecture.	CO4	2 3	8 6
	OR			
8	a) Differentiate between PoW Consensus vs BFT Consensus b) What are the function modifiers in solidity? How is blockchain distributed ledger different from the traditional ledger?	CO4	2 2	7 7
9	a) What is Identity? Explain Centralized Digital Identity b) Explain how blockchain can be used in financial services	CO5	2 2	7 7
	OR			
10	a) Write about decentralized identifiers b) Discuss problem with interoperability in blockchain using an example	CO5	2 2	6 8

*****All The Best*****